

1.

```
5 Scanner sc = new Scanner(System.in);
6
7 int arr[] = {20, 40, 60, 80};
8
9 int index = sc.nextInt();
10 int divisor = sc.nextInt();
11
12 try {
13     int value = arr[index];
14     int result = value / divisor;
15     System.out.println(result);
16 } catch (ArrayIndexOutOfBoundsException e) {
17     System.out.println("Invalid Index");
18 } catch (ArithmeticException e) {
19     System.out.println("Division by Zero");
20 }
21 }
22 }
```

Output

12

Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

2.

```
4
5 static int divide(int a, int b) {
6     return a / b;
7 }
8
9 static void display(int x) {
10     System.out.println("Answer = " + x);
11 }
12
13 public static void main(String[] args) {
14     Scanner sc = new Scanner(System.in);
15
16     int a = sc.nextInt();
17     int b = sc.nextInt();
18
19     try {
20         int result = divide(a, b);
21         display(result);
22     } catch (ArithmeticException e) {
23     }
```

Output

```
Answer = 4
Execution Completed
```

Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

3.

```
5     Scanner sc = new Scanner(System.in);
6
7     int arr[] = {12, 24, 36, 48};
8
9     String number = sc.next();
10
11     try {
12         int index = Integer.parseInt(number);
13         System.out.println(arr[index]);
14     } catch (NumberFormatException e) {
15         System.out.println("Invalid Number");
16     } catch (ArrayIndexOutOfBoundsException e) {
17         System.out.println("Invalid Array Index");
18     }
19
20     sc.close();
21 }
22 }
```

Output

```
36
```

Test Cases

Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

10.00 / 10 marks

4.

```
6     try {
7         return a / b;
8     } finally {
9         System.out.println("Resources Released");
10    }
11 }
12
13 public static void main(String[] args) {
14     Scanner sc = new Scanner(System.in);
15
16     int a = sc.nextInt();
17     int b = sc.nextInt();
18
19     try {
20         System.out.println(calculate(a, b));
21     } catch (ArithmeticException e) {
22         System.out.println("Division by Zero");
23     }
24 }
```

Output

```
Resources Released
4
```

Test Cases

Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

10.00 / 10 marks

5.

```
4 public static void main(String[] args) {  
5     Scanner sc = new Scanner(System.in);  
6  
7     String name = sc.nextLine();  
8  
9     if (name.equalsIgnoreCase("null")) {  
10         name = null;  
11     }  
12  
13     if (name != null) {  
14         System.out.println("Length = " + name.length());  
15     } else {  
16         System.out.println("String is Null");  
17     }  
18  
19     sc.close();  
20 }  
21 }
```

Output

```
Exception in thread "main"  
java.util.NoSuchElementException: No line found  
    at  
    java.base/java.util.Scanner.  
    nextLine(Scanner.java:16
```

Test Cases

Test Case 1	✓
Test Case 2	✓
Test Case 3	✓

10.00 / 10 marks